

Assignment 2
Hyper turning Parameter

1. Support Vector Machine

S.NO	Hyper Parameter	Linear	RBF	Poly	Sigmoid
1	C10	-0.5032	-0.5080	-0.527	-0.5093
2	C100	0.5262	0.5013	0.5040	0.5056
3	C500	0.6803	0.6397	0.815	0.4638
4	C1000	0.6794	0.7915	0.8519	0.1842
5	C2000	0.7013	0.8460	0.8573	-0.5786
6	C3000	0.7312	0.8609	0.8577	-2.0119

The r_score is RBF in C 3000 is a best model

2. Decision Tree Regression

S.No	Criteion	Splitter	R_score
1	friedman	best	0.90
2	friedman	random	0.21
3	absolute error	best	0.92
4	absolute error	random	0.75
5	poisson	best	0.92
6	poisson	random	0.62
7	squared_root	best	0.90
8	squared_root	random	0.88
9	default	default	0.92

The r_score is deflaut is a best model 0.92

3. Random Forest

S.NO	Criterion	Max Features	N_Estimators	R Value
1	Mse	Auto	10	0.9216
2	Mse	Auto	50	0.9901
3	Mse	Sqrt	10	0.9271
4	Mse	Sqrt	100	0.99722
5	Mse	Log2	10	0.9271
6	Mse	Log2	100	0.9122
7	Mae	Auto	10	0.9265
8	Mae	Auto	100	0.9992
9	Mae	Sqrt	10	0.9232
10	Mae	Sqrt	100	0.9964
11	Mae	Log2	10	0.9232
12	Mae	Log2	100	0.9964